RetailDWH: Grain, Keys, and SCD Design

Validation and Analytical Questions (Queries in File dim_fact_Data_Quality_Validations)

1. **How many source orders have a Completed status? 161**

```
2.  SELECT COUNT(DISTINCT OrderID) fact_orders_count
FROM   [fact].[FactSales];

SELECT COUNT(DISTINCT [OrderLines].[OrderID]) stg_order_count
FROM   [stg].[Orders]
       INNER JOIN
       [stg].[OrderLines]
       ON [Orders].[OrderID] = [OrderLines].[OrderID]
WHERE  OrderStatus = N'Completed'
       AND EXISTS (SELECT *
                   FROM   [stg].[ProductCatalog]
                   WHERE  [OrderLines].[ProductID] = [ProductCatalog].[ProductID]);
```

	fact_orders_count
1	161

	stg_order_count
1	161

3. **How many valid order lines remain after the product lookup? 430**

```
SELECT COUNT(*) AS fact_OrderLines
FROM   fact.FactSales;

SELECT COUNT(*) AS stg_valid_OrderLines
FROM   stg.Orders AS o
       INNER JOIN
       stg.OrderLines AS ol
       ON o.OrderID = ol.OrderID
WHERE  o.OrderStatus = N'Completed'
       AND EXISTS (SELECT 1
                   FROM   stg.ProductCatalog AS p
                   WHERE  p.ProductID = ol.ProductID);
```

	fact_OrderLines
1	430

	stg_valid_OrderLines
1	430

4. **Does COUNT(DISTINCT SourceOrderLineID) equal the total row count in FactSales?**

```
SELECT COUNT(DISTINCT SourceOrderLineID) AS DistinctSourceOrderLines,
       COUNT(*) AS TotalFactRows,
       CASE WHEN COUNT(DISTINCT SourceOrderLineID) = COUNT(*) THEN 'Yes' ELSE 'No' END AS
CountsMatch
FROM   fact.FactSales;
```

	DistinctSourceOrderLines	TotalFactRows	CountsMatch
1	430	430	Yes

5. **What is the monthly revenue by brand and store?**

```sql
SELECT
MonthName,
MonthNumber,
Brand,
StoreName,
SUM(NetSales) AS monthly_revenue
FROM fact.FactSales
INNER JOIN dim.DimDate
    ON FactSales.DateKey = DimDate.DateKey
INNER JOIN dim.DimProduct
    ON FactSales.ProductKey = DimProduct.ProductKey
INNER JOIN dim.DimStore
    ON FactSales.StoreKey = DimStore.StoreKey
GROUP BY
    MonthName,
    MonthNumber,
    Brand,
    StoreName
ORDER BY
   MonthNumber,
   Brand,
   StoreName;
```

	MonthName	MonthNumber	Brand	StoreName	monthly_revenue
1	July	7	Nest	Alex Store	894.82
2	July	7	Nest	Dokki Store	245.28
3	July	7	Nest	Downtown Store	367.70
4	July	7	Nest	Ismailia Store	1082.91
5	July	7	Nest	Mansoura Store	1698.29
6	July	7	Nest	Nasr City Store	44.77
7	July	7	Urban	Alex Store	599.82
8	July	7	Urban	Dokki Store	1965.43
9	July	7	Urban	Downtown Store	965.04
10	July	7	Urban	Ismailia Store	1675.21
11	July	7	Urban	Mansoura Store	4051.05
12	July	7	Urban	Nasr City Store	275.76
13	July	7	Volt	Alex Store	498.44
14	July	7	Volt	Dokki Store	1076.40
15	July	7	Volt	Downtown Store	1203.28
16	July	7	Volt	Ismailia Store	274.00
17	July	7	Volt	Mansoura Store	1537.79
18	July	7	Volt	Nasr City Store	637.93
19	July	7	WorkPro	Alex Store	779.52
20	July	7	WorkPro	Dokki Store	1478.02
21	July	7	WorkPro	Downtown Store	2425.84
22	July	7	WorkPro	Ismailia Store	1021.97
23	July	7	WorkPro	Mansoura Store	1611.64
24	July	7	WorkPro	Nasr City Store	2662.86

1. Fact Table Grain

The grain of FactSales is **one row per order line**. Each record represents an individual product line within an order, sourced from stg.Orders and stg.OrderLines. Only Completed orders contribute to completed sales revenue; Pending and Cancelled orders are excluded.

OrderID is retained in FactSales as a **degenerate dimension**, allowing orders to be identified without creating a separate DimOrder table.

2. Business Keys and Surrogate Keys

Business keys identify records in the source system, while surrogate keys provide warehouse-specific identifiers for dimension records.

- **DimCustomer:** CustomerID is the business key; CustomerKey is the surrogate key.

- **DimStore:** StoreID is the business key; StoreKey is the surrogate key.
- **DimProduct:** ProductID is the business key; ProductKey is the surrogate key. Because SCD Type 2 is applied, one ProductID can have multiple ProductKey values representing historical versions.
- **DimDate:** DateKey identifies the date dimension record and is used to link facts to their corresponding dates.

FactSales stores the appropriate dimension surrogate keys, which are resolved using source business IDs and the order date. Four foreign key constraints enforce referential integrity between the fact and dimension tables.

3. Slowly Changing Dimension (SCD) Choices

- **DimCustomer — SCD Type 1:** Existing records are updated with the latest values without preserving historical versions.
- **DimStore — SCD Type 1:** Existing records are updated with the latest values without preserving historical versions.
- **DimProduct — SCD Type 2:** Historical product versions are preserved, particularly when product costs change. This supports accurate historical profit calculations by associating each order line with the product cost valid at the order date.
- **DimDate:** A date dimension used for consistent time-based analysis; SCD processing is not required.

For DimProduct, the effective-date interval follows StartDate <= OrderDate < EndDate. The first version of a product uses the default start date 1900-01-01. When a tracked change occurs, the previous version is closed using the new version's effective date, and the new version receives that date as its StartDate. The current version uses EndDate = NULL. A CHECK constraint enforces the documented date-range rule.

4. Missing Dimension Members

Unknown members with key 0 are maintained in DimCustomer and DimStore for missing or unavailable references. Inferred members can be created for other missing customer or store business keys and identified using IsInferredMember = 1.

If an order line references a product that does not exist in DimProduct, it is redirected to an output file for investigation rather than being assigned an arbitrary ProductKey. Inferred member handling for DimProduct is currently disabled.

5. Loading Strategy

The warehouse dimensions and fact table use incremental loading based on the last successful watermark. Dimensions are processed before FactSales so that valid surrogate keys are available and referential integrity can be checked before fact loading.

6. Bonus Features and Additional Implementations

- **Incremental Loading:** Implemented weekly incremental loading for the staging layer and incremental loading for dimensions and facts, using a Watermarks table to track the last successful load point and process only new or modified records.
- **Dual Staging Load Strategies:** Implemented monthly Overwrite loading using TRUNCATE and INSERT, alongside weekly Incremental loading.
- **SSIS Master–Child Orchestration:** Designed a master package to execute separate child packages. A failure in one child package does not prevent the remaining packages from executing.
- **Batch and Table-Level Logging:** Implemented batch tracking with start/end timestamps and overall execution status, alongside logging of table operations such as TRUNCATE, INSERT, and UPDATE, including their outcomes and associated batch IDs.
- **Automated Error Handling:** Added execution-status tracking so failed operations are logged and the batch is marked as Failed when a required operation fails.
- **Inferred Member Handling:** Added a strategy for handling missing customer and store dimension records using inferred members flagged with IsInferredMember = 1, allowing their attributes to be completed when source data becomes available. Product inferred-member handling is currently disabled.
- **Data Quality and Referential Integrity:** Added validation checks before warehouse loading to identify invalid data and missing dimension references.
- **Product History Management:** Implemented SCD Type 2 for DimProduct, including effective-date handling and a date-range CHECK constraint to support historical product-cost analysis.
- **Profit Calculation:** Implemented a derived Profit column in FactSales, calculated as NetSales - (Quantity × UnitCost). The historical UnitCost is resolved using the appropriate DimProduct SCD Type 2 version based on the order date, ensuring accurate profit calculations using the cost valid at the time of the sale.